

Marketing Campaign and Customer Insights Analysis

Results

Insights

Optimization

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Tools Used:

SQL and Tableau



Project Overview

Marketing campaigns generate large amounts of customer data, from demographic information to campaign engagement and spending habits. When analyzed effectively, it helps businesses understand how to improve campaign targeting and increase return on marketing investment.

Problem:

A marketing team has customer data (demographics, spending, campaign responses) but no clear read on who their best customers are, what drives revenue, or whether their past campaigns actually worked. This project turns raw customer data into a decision-ready view of customer value and campaign performance.



Key Objectives

01

Who are the highest-value customers, and what do they have in common?

02

Which products and purchase channels drive the most revenue?

03

How effective were the past marketing campaigns?

04

How has customer revenue grown over time?

Analysis Techniques



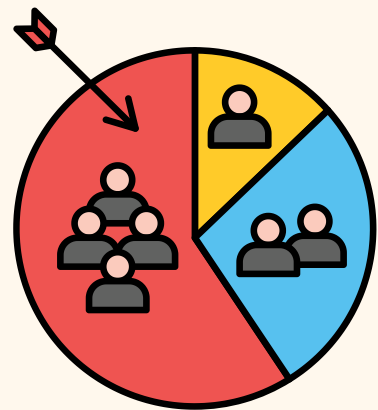
Data Cleaning and Preparation

- Duplicate detection
- Missing data assessment
- Categorical standardization



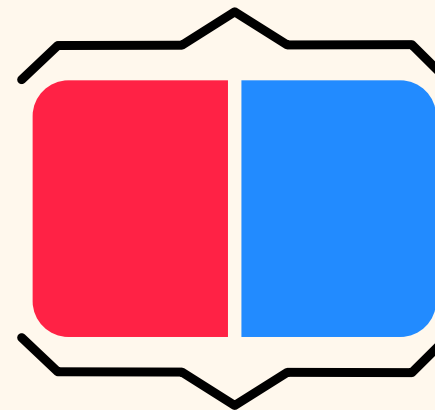
Descriptive statistics

- Central tendency
- Frequency distribution
- Totals/Sums



Segmentation Analysis

- Grouping customers by different factors
- Quartile segmentation



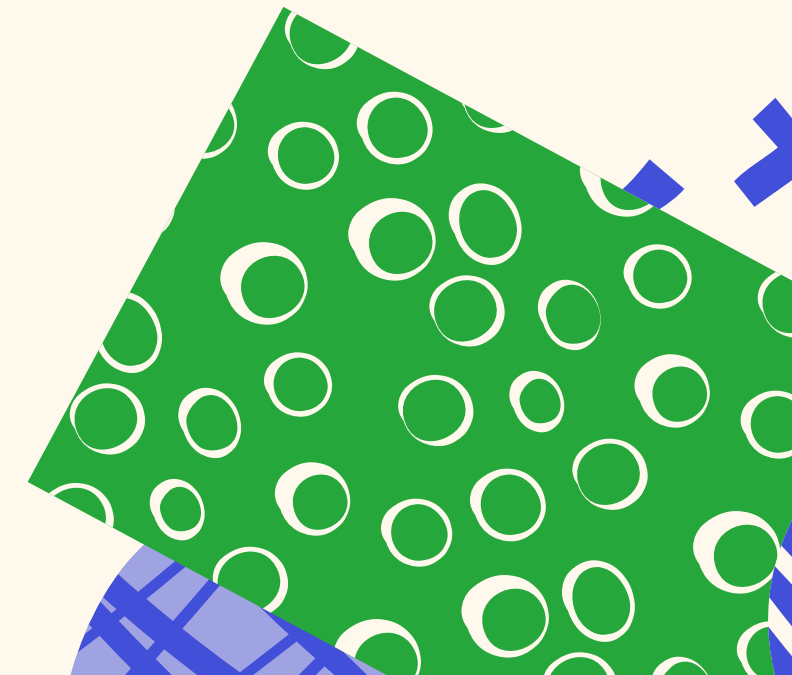
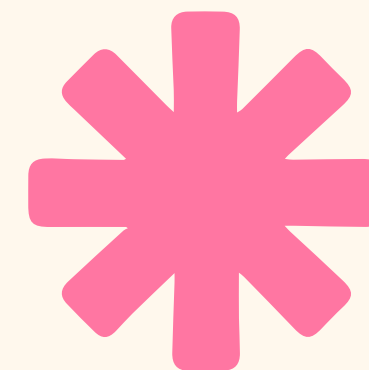
Comparative Analysis

- Responders vs. non-responders
- Campaign to campaign comparison



Trend Analysis

- Customer growth over time
- Revenue growth over time

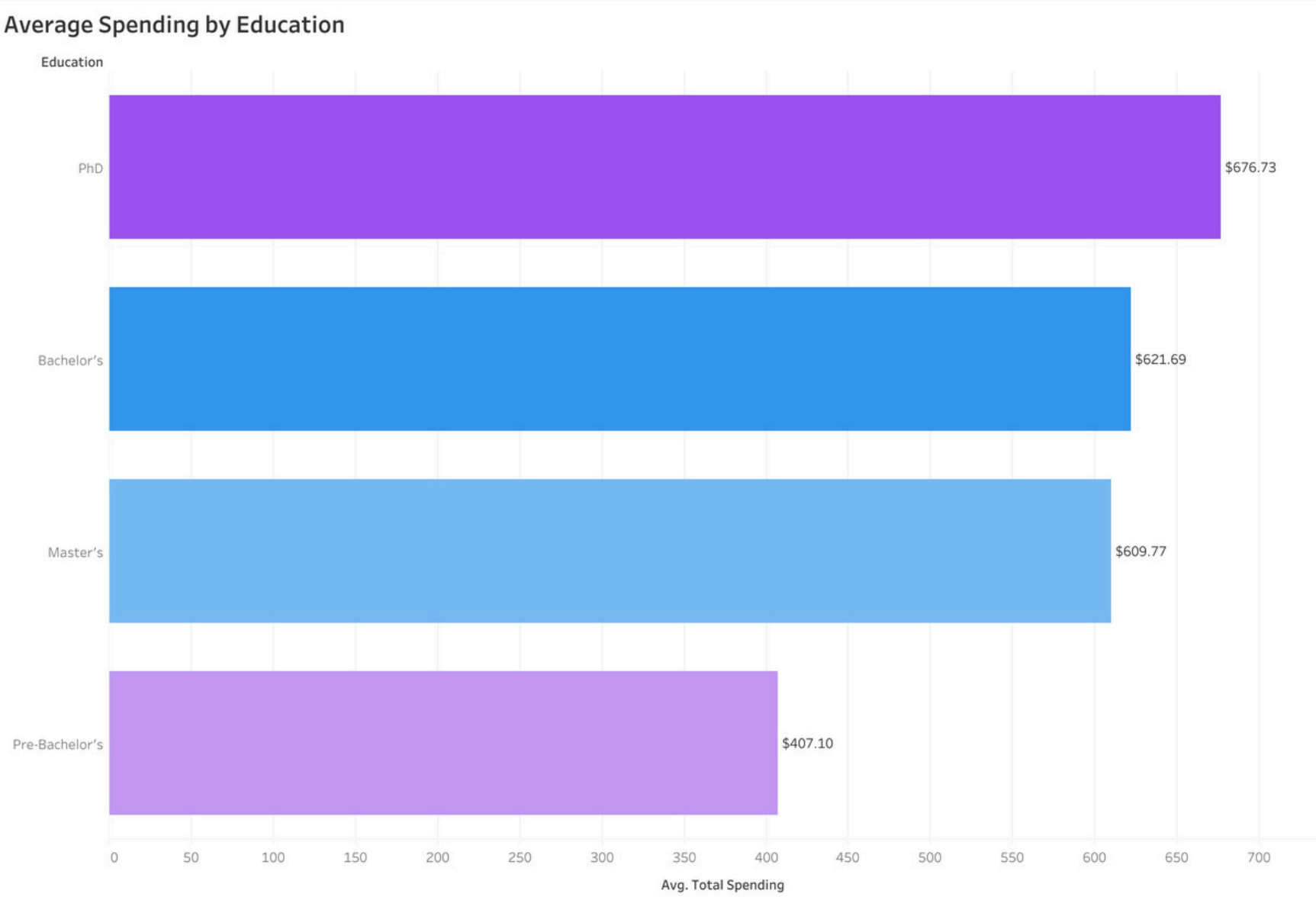


Customer Insights

Table 1: Customer Distribution by Education

Education Level	Total Customer
Pre-Bachelor's	11.46%
Bachelor's	50.36%
Master's	16.47%
PhD	21.71%

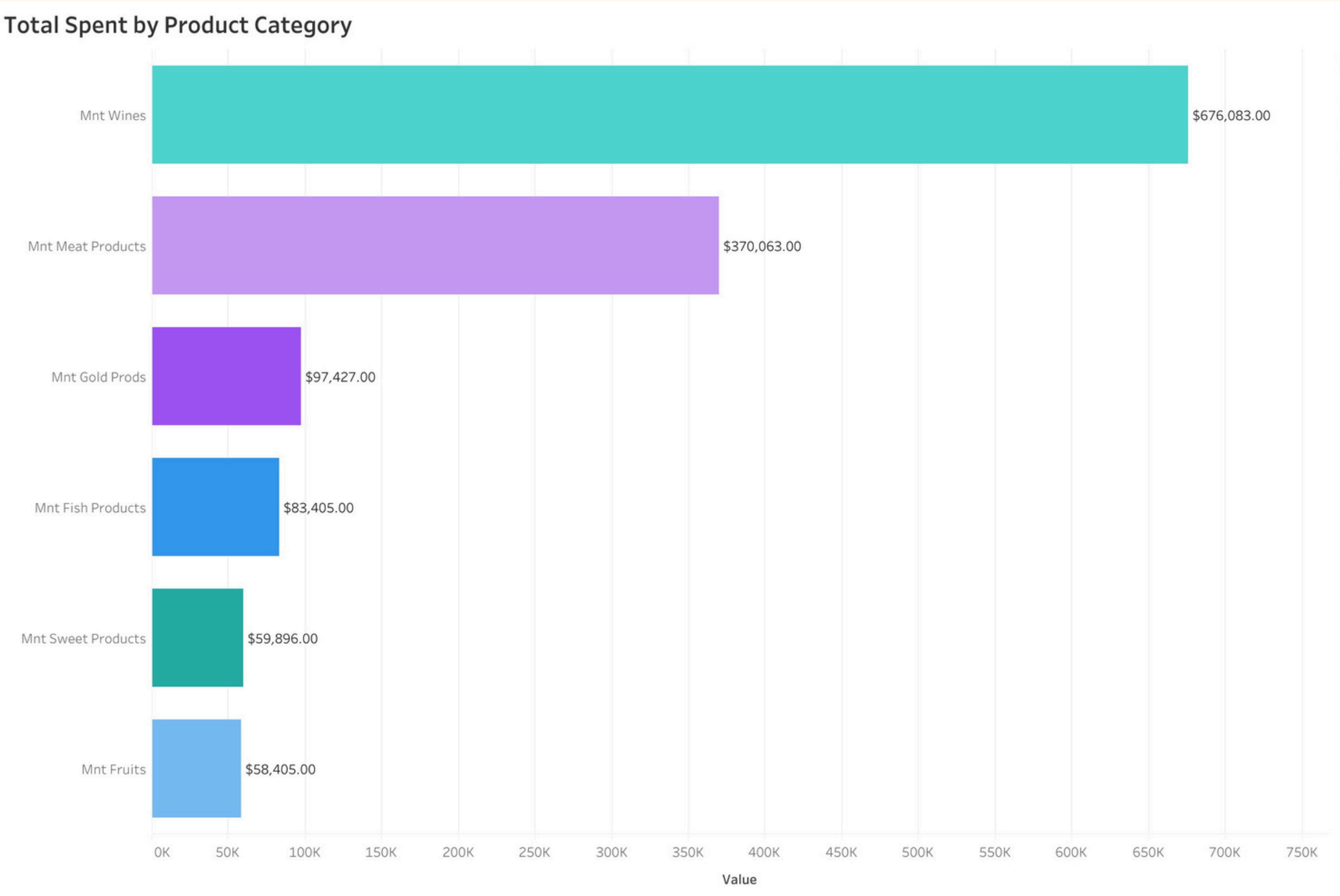
Figure 1: Bar Chart – Average Spending by Education



Customers are heavily concentrated in the **Bachelor's segment** (50.36% of the base), with **PhD** (21.71%), **Master's** (16.47%), and **Pre-Bachelor's** (11.46%) making up the rest. But **average spending** runs in almost the **opposite direction of segment size**: **PhD customers** spend **the most** per person (\$676.73), while the largest segment, Bachelor's, **spends less individually** (\$621.69), and Pre-Bachelor's trails well behind at \$407.10.

Product Performance

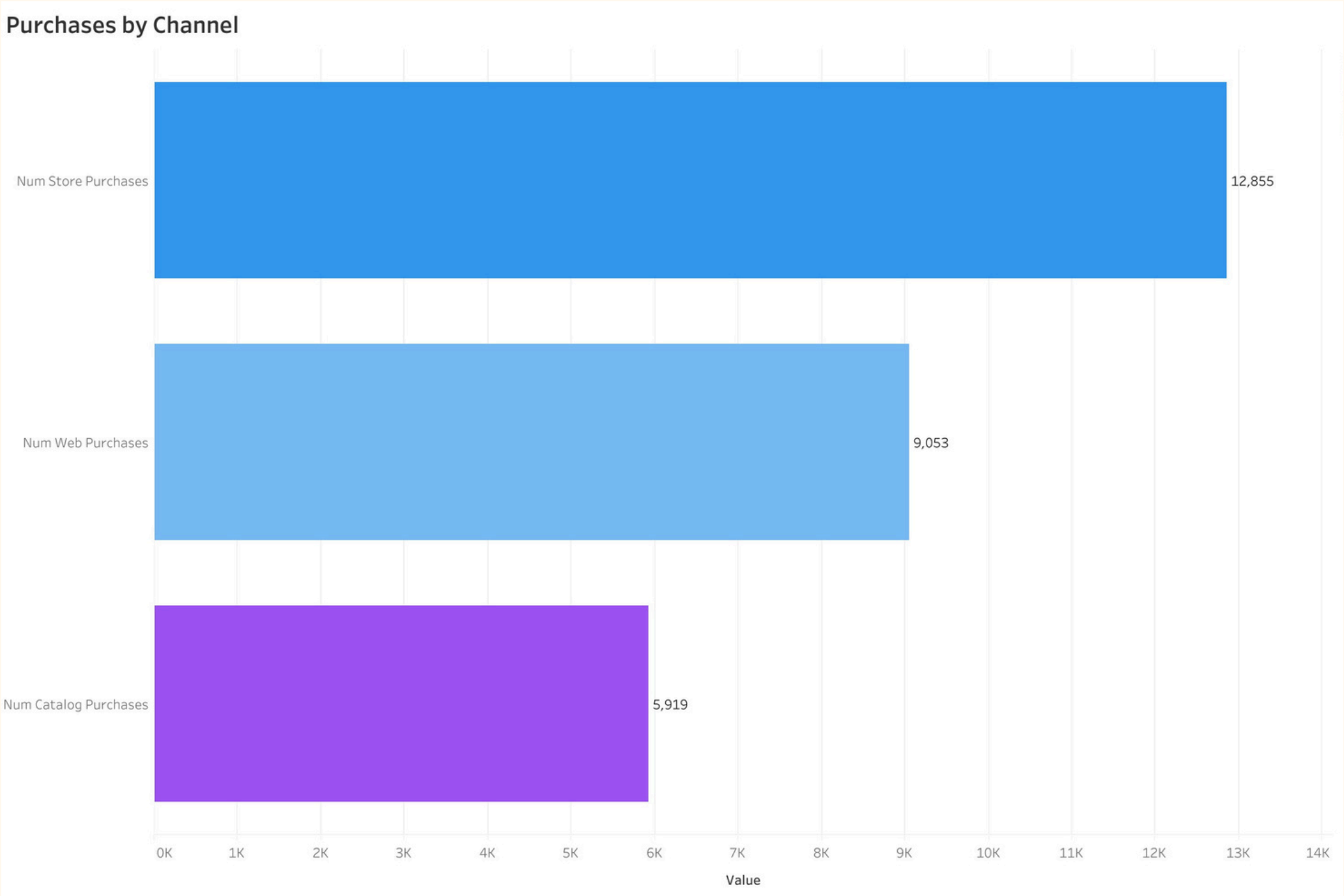
Figure 2: Total Spending by Product Category



Wine dominates the product mix (\$676,083), **nearly double** the next category (Meat, \$370,063). Gold, Fish, Sweet, and Fruit products all fall well below \$100K each, making them minor contributors by comparison.

Channel Performance

Figure 3: Product Purchases by Channel



Store purchases **lead** (12,855), ahead of **Web** (9,053) and **Catalog**, which trails distinctly (5,919).

Campaign Performance

Figure 4: Total Customers Accept the Offers through Each Campaign

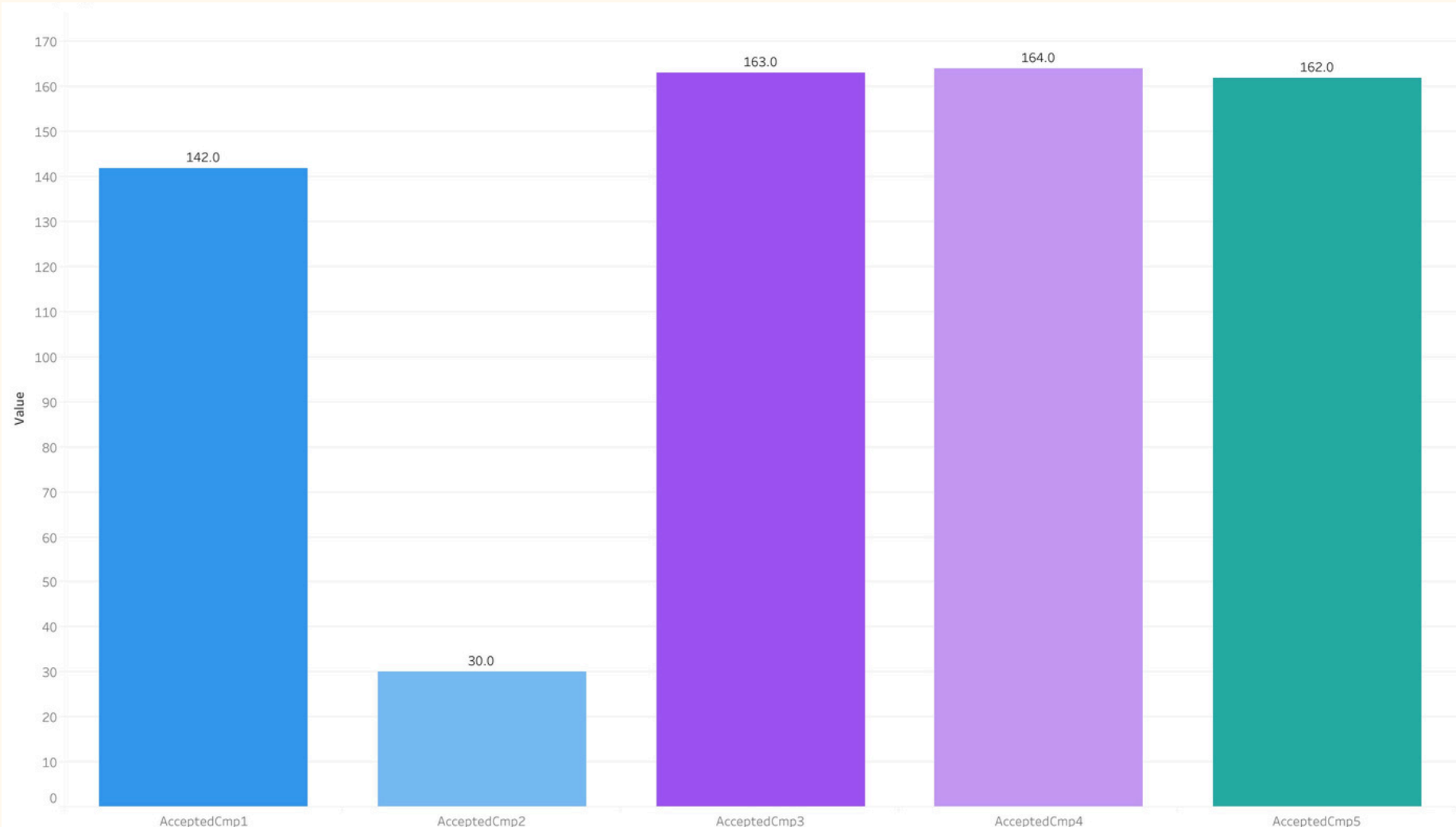


Table 2: Value of Campaign Responders

Factor	Result
Customers	333
Average Spending	\$985.66
Revenue	\$328,225

Figure 4 shows **Campaigns 3, 4, and 5 performing similarly** (162–164 acceptances), **Campaign 1** slightly behind (142), and **Campaign 2** far below all of them (30). Table 2 shows what those responders are actually worth: the **333 customers who accepted a campaign** (15% of the base) spent **\$985.66 on average** and **generated \$328,225 in total revenue**, meaning this small responder group is disproportionately valuable relative to its size.

Business’s Performance

Figure 5: Revenue Trend

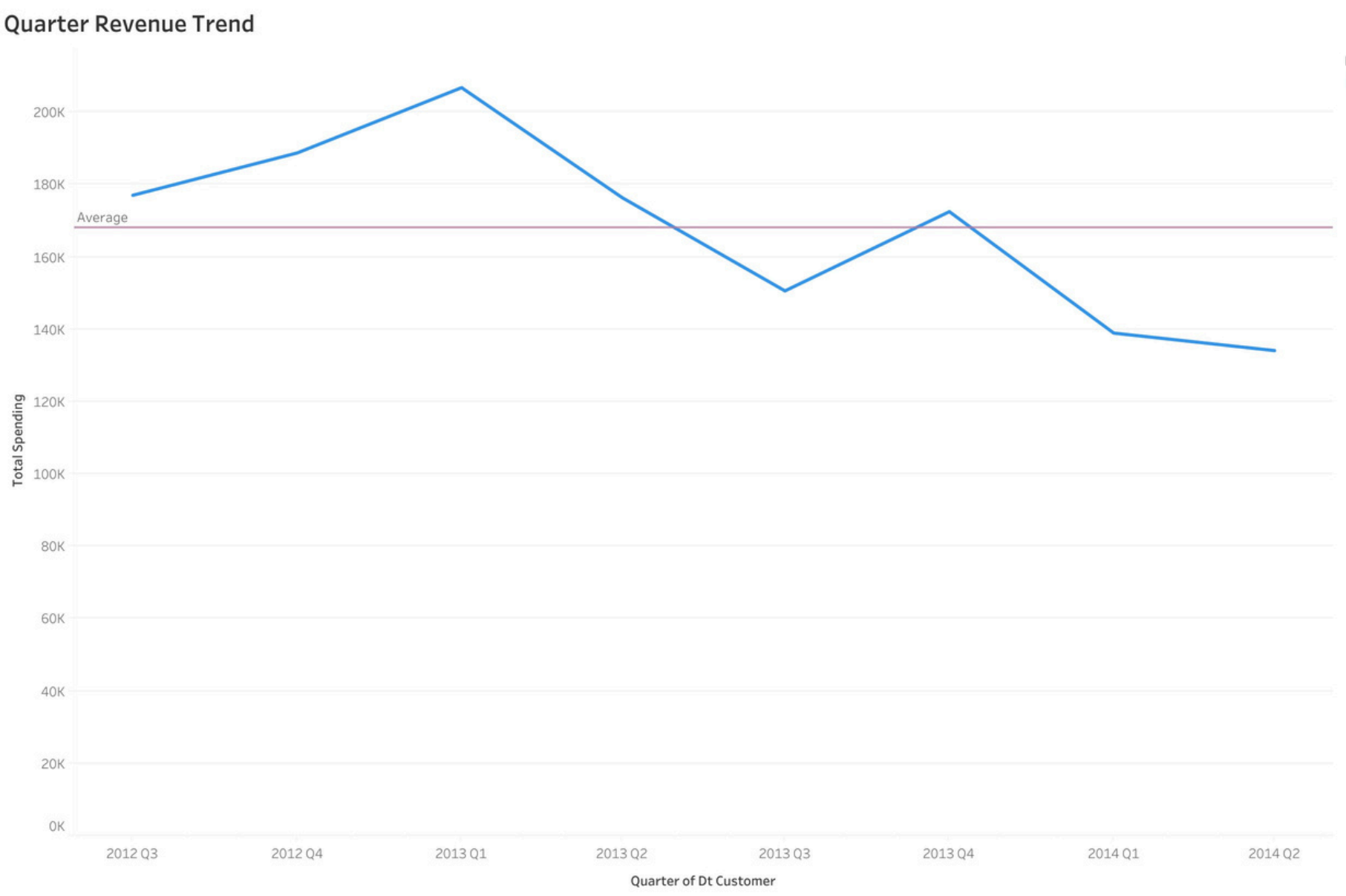
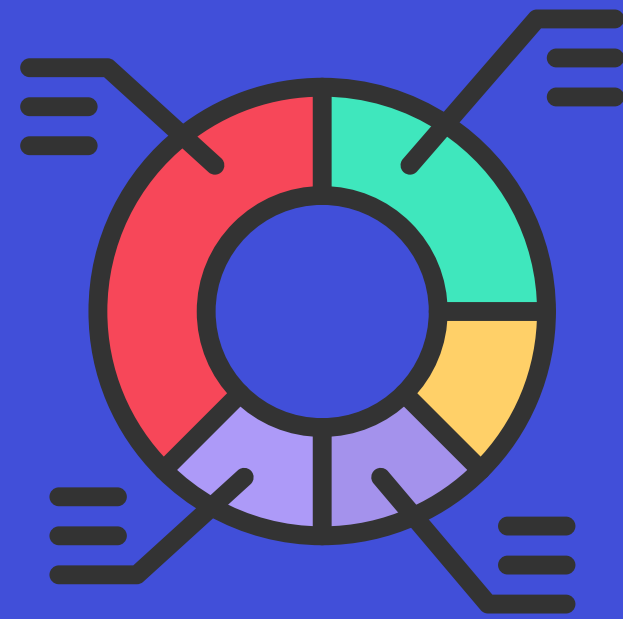


Table 3: Business Growth Over the Year

Year	Total Customer	Total Revenue
2012	490	\$365,833
2013	1173	\$706,357
2014	553	\$273,809

Customer acquisition and revenue both **peaked in 2013** (1,173 new customers, \$706,357), after growing from 2012 (490 customers, \$365,833), then **dropped sharply in 2014** (553 customers, \$273,809).

Key Findings



Segment size and segment value move in opposite directions.



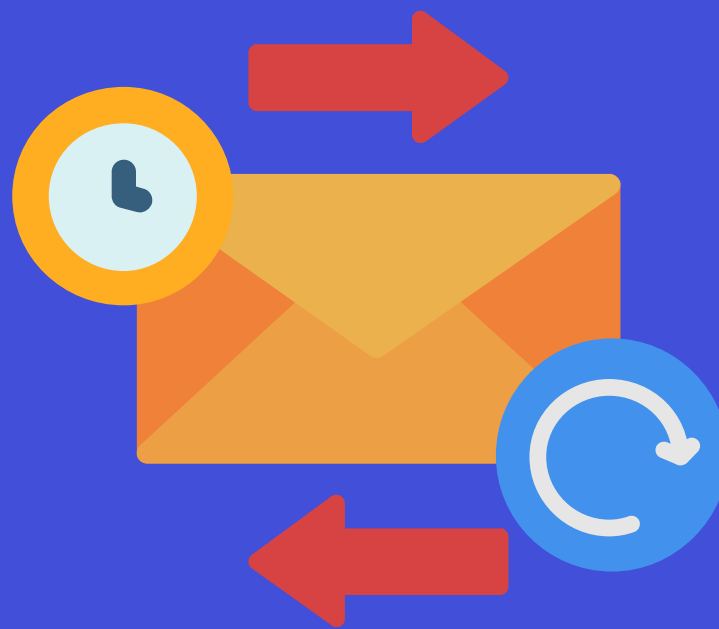
Wine is the single biggest revenue driver.



Store remains the primary purchase channel



Campaign 2 is a clear underperformer.



Campaign responders are a small but high-value group.



2013 was the acquisition and revenue peak.

Recommendations

Segment marketing by value, not size.

PhD customers spend the most per person and justify premium offers, while Bachelor's customers, the largest segment, respond better to volume-driven retention campaigns.

Protect and cross-sell around Wine.

It drives more revenue than every other product category combined, making it a strong anchor for bundling promotions with underperforming categories like Fruit and Sweets.

Keep prioritizing the Store channel.

Store purchases outperform Web and Catalog combined, so digital investment shouldn't come at the expense of in-store experience.

Pause or redesign Campaign 2.

Its acceptance rate is 4–5x lower than every other campaign, reallocating that budget toward Campaigns 3–5 is likely to yield a better return.

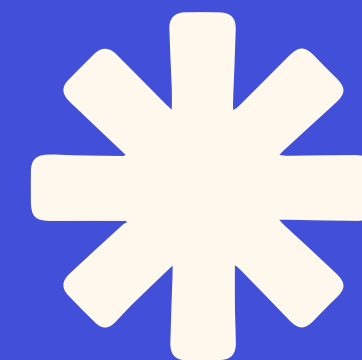
Prioritize responder retention over new acquisition.

The 333 customers who responded to campaigns already generate \$328K in revenue at 82% higher spend than non-responders. Deepening engagement with this group is likely more cost-effective than chasing new customers.

Conclusions



The data reveals a clear overall picture: **the business has a small but high-value customer segment among PhD customers**, while the larger Bachelor's segment generates the highest total revenue despite having lower value per customer. The business also shows a **strong reliance on wine products**, a **Store-led purchasing strategy**, and a relatively small group of campaign responders who contribute disproportionately to overall value.



The two most actionable findings are **Campaign 2's underperformance** and the high value generated by the 333 campaign responders. Both findings are specific, measurable, and can directly inform the next quarter's marketing strategy.

Additionally, the apparent 2014 revenue drop, once investigated, turned out to be a **partial-year data artifact rather than a real decline**. Thus, a reminder that a sharp swing in the data is worth verifying before it's reported as a trend, since 2014 was likely tracking in line with 2013's growth rather than falling off.

[View Dashboard](#)

Thank You!

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